A Wide-Tuning, Low-K_{VCO}-Variation 5 GHz Complementary LC-VCO with a 3-bit Switched-Capacitor Bank in 180 nm CMOS

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***Abstract*—This report presents the design, simulation and characterization of a complementary cross-coupled LC voltage-controlled oscillator (VCO) with a 3-bit switched-capacitor bank and small oscillator-gain variation, implemented in the TSMC 0.18 μm RF CMOS process for the 5 GHz wireless local-area network band. The design covers 5.226–5.855 GHz, a fractional tuning range of 11.35 %, across eight frequency bands that tile the range continuously with no dead zones. The oscillator gain K_{VCO} varies by less than 21 MHz/V over the entire control range (±11.5 % about its mean) and remains below 105 MHz/V everywhere. A phase noise of -116.4 dBc/Hz at a 1 MHz offset is achieved with a differential output swing of ±1.25 V while drawing 3.57 mA from a 1.8 V supply, giving a figure of merit of -182.7 dB. A preliminary layout of the oscillator core is also presented; it is unverified and is included to document the floorplanning strategy only.**

***Index Terms*—Voltage-controlled oscillator, LC tank, switched-capacitor bank, phase noise, CMOS, WLAN, figure of merit.**

I. Introduction

A. Background

The voltage-controlled oscillator is the frequency-defining element of a wireless transceiver. Embedded within a phase-locked loop (PLL), it generates the local-oscillator signal that drives the up- and down-conversion mixers. Its phase noise is one of the principal limits on receiver performance, since oscillator sidebands reciprocally mix unwanted adjacent-channel energy into the intermediate frequency. Phase noise is therefore integral to the overall design goal of the VCO.

The 5 GHz unlicensed national information infrastructure (U-NII) band, used by IEEE 802.11a/n/ac/ax wireless local-area networks, spans approximately 5.15–5.85 GHz. Covering this band with a single oscillator requires a wide tuning range. This requirement conflicts with the desire for a low oscillator gain, K_{VCO}. If a single varactor is asked to cover the entire band, the frequency must change steeply with the tuning voltage, and any noise present on the tuning line or on the supply is then converted efficiently into frequency, creating phase modulation.

The standard resolution of this conflict is to divide the tuning task between two mechanisms. A switched-capacitor bank provides coarse, discrete frequency bands, while a varactor performs only fine, continuous tuning within each band. The total tuning range is then the sum of many narrow bands rather than the range of a single wide-swinging varactor, and the oscillator gain within each band can be kept low. A low and weakly varying K_{VCO} also stabilizes the loop bandwidth of the enclosing PLL, since the loop gain is directly proportional to K_{VCO} [1].

B. Objectives

The objectives of this work were: (i) to design a complementary LC-VCO in a 0.18 μm CMOS technology capable of covering the 5 GHz WLAN band with continuous, gap-free frequency coverage; (ii) to design and dimension a switched-capacitor bank with adequate overlap margin; (iii) to characterize the oscillator fully at the schematic level, including tuning range, oscillator gain, phase noise and figure of merit; and (iv) to compare the resulting performance with recently published oscillators in comparable technologies. The intended outcome is a validated oscillator core suitable for subsequent layout and PLL integration.

II. Design Methodology

A. Design Environment and Technology

The oscillator was designed and simulated in Keysight Advanced Design System (ADS) using the TSMC 0.18 μm RF CMOS process design kit. All devices — the spiral inductor, the accumulation-mode MOS varactors, the metal–insulator–metal capacitors of the bank and the poly resistors — were taken from the foundry RF library, so that device parasitics and quality factors are those of the target process rather than idealized elements.

B. Oscillator Core

The oscillator uses a complementary cross-coupled topology, in which a PMOS pair and an NMOS pair jointly generate the negative resistance that compensates the loss of the resonant tank. Relative to an NMOS-only implementation, the complementary structure reuses the bias current across both device types, so that a given current yields a larger total transconductance. It also produces a more symmetric output waveform. This waveform symmetry suppresses the DC term of the impulse sensitivity function, and it is this term that

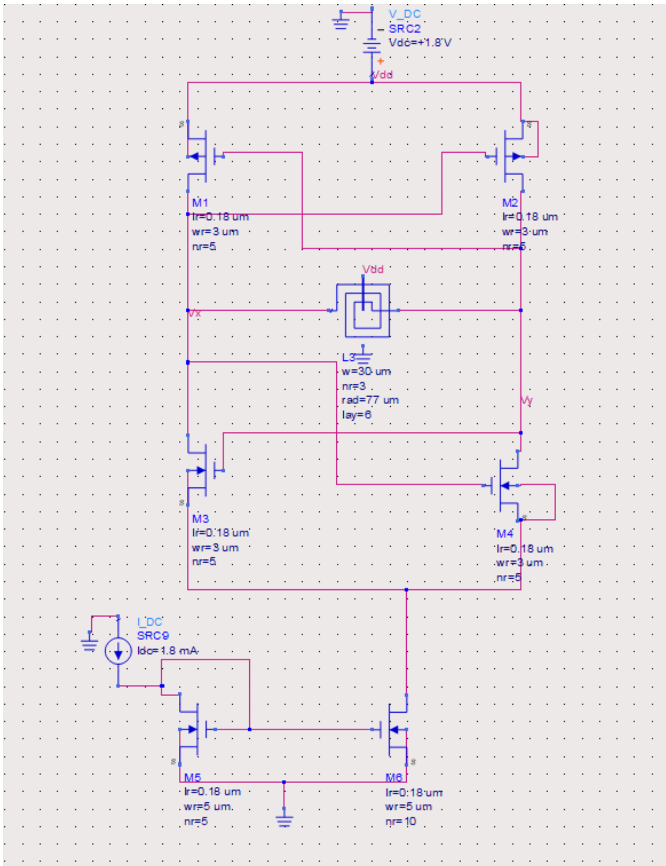

Fig. 1. Schematic of the complementary cross-coupled LC-VCO, showing the PMOS and NMOS cross-coupled pairs, the centre-tapped spiral inductor and the NMOS tail current mirror.

governs the upconversion of low-frequency flicker noise into close-in phase noise [2]. Bias current is supplied by an NMOS current mirror. The core schematic is shown in Fig. 1.

Sustained oscillation requires that the negative resistance of the active pairs overcome the loss of the tank. Expressed as a condition on the transconductance and the equivalent parallel tank resistance R_p, startup requires

$$g_m R_p > 1, \tag{1}$$

with a margin of two to three times applied in practice to guarantee reliable startup across process corners. Since $R_p = Q\,\omega L$, a tank of low inductance or low quality factor presents a low impedance, demands a correspondingly larger transconductance, and yields a smaller oscillation amplitude for a given bias current. The choice of inductor is therefore consequential, and a three-turn symmetric spiral with a centre tap to the supply was selected to obtain sufficient inductance at the target frequency.

C. Frequency Tuning Architecture

The resonant tank comprises the spiral inductor, a pair of accumulation-mode varactors providing continuous tuning, and a 3-bit binary-weighted switched-capacitor bank providing coarse band selection (Fig. 2). The oscillation frequency is

$$f_0 = \frac{1}{2\pi\sqrt{L\,C_{\text{total}}}}, \tag{2}$$

where C_{total} is the sum of the varactor capacitance, the selected bank capacitance and the parasitic capacitance of the active devices. Differentiating with respect to capacitance gives the tuning sensitivity

$$\frac{df}{dC} = -\frac{f}{2\,C_{\text{total}}}. \tag{3}$$

Equation (3) has two consequences that shaped the design. First, the frequency step produced by a given bank capacitor is not constant across the tuning range: as capacitance is switched in, C_{total} rises and f falls, and both effects reduce the frequency shift produced by each additional capacitance. The bank cannot therefore be dimensioned from a single nominal sensitivity. Second, and by the same reasoning, the oscillator gain K_{VCO} — the frequency change produced by the varactor per volt of tuning — must fall as the oscillator is tuned to lower bands. Both effects were observed in simulation and are reported in Section III.

D. Dimensioning the Switched-Capacitor Bank

The bank capacitors are drawn in a binary $1:2:4$ ratio, nominally 43 fF, 85 fF and 135 fF, each placed in series with an NMOS switch, and every branch is mirrored on both tank nodes to preserve differential symmetry. A practical difficulty arises at the node between the capacitor and its switch: when the switch is off, this node has no path to any DC reference, being blocked by the capacitor on one side and by the off transistor on the other. A large pull-up resistor from this intermediate node to the supply resolves the problem, providing a DC reference of sufficiently high impedance that it does not load the tank at the frequency of oscillation [3].

The bank was dimensioned against the no-dead-zone constraint. The frequency step produced by the least significant bit must be smaller than the narrowest varactor tuning span, so that adjacent bands overlap:

$$\Delta f_{\text{step}} < \Delta f_{\text{varactor}}, \tag{4}$$

with the difference between the two constituting the overlap margin. Because the varactor span itself shrinks at the lower bands, the binding case is the lowest band, and the bank was sized against that worst case rather than against the nominal sensitivity.

E. Figure of Merit

For comparison with published oscillators, the standard figure of merit is used:

$$\text{FoM} = \mathcal{L}\{\Delta f\} - 20\log_{10}\left(\frac{f_0}{\Delta f}\right) + 10\log_{10}\left(\frac{P_{\text{DC}}}{1\,\text{mW}}\right), \tag{5}$$

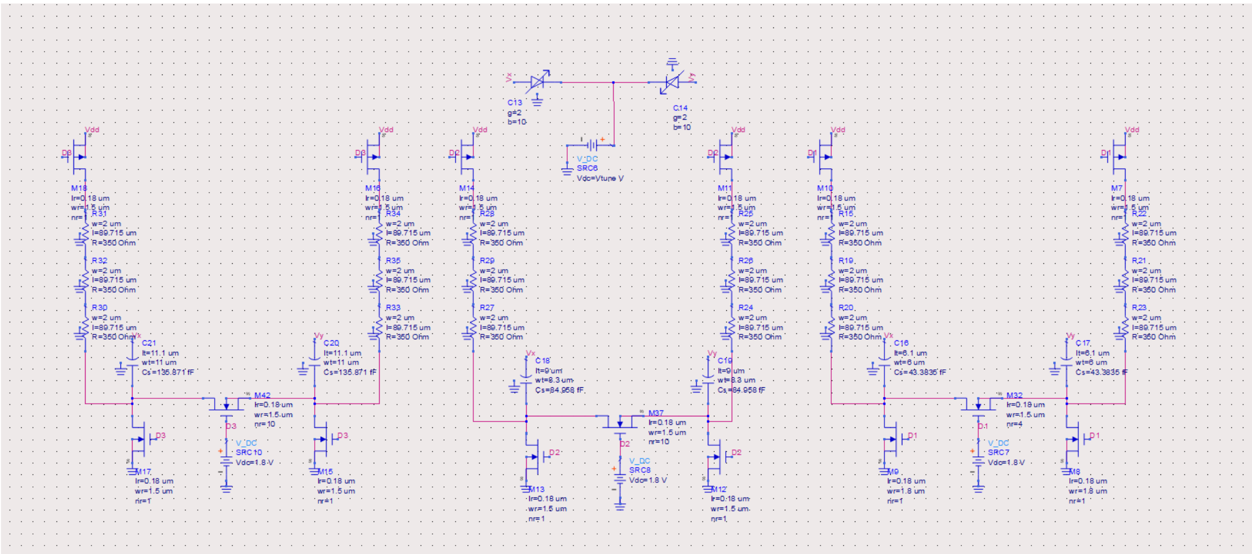

Fig. 2. Schematic of the continuous-tuning varactor pair and the 3-bit binary-weighted switched-capacitor bank. Every branch is mirrored on both tank nodes to preserve differential symmetry, and each intermediate node is biased through a large pull-up resistor.

where $\mathcal{L}\{\Delta f\}$ is the single-sideband phase noise at offset Δf from the carrier f_0, and P_{DC} is the DC power consumed by the oscillator core. A tuning-range-normalized variant,

$$\mathrm{FoM_T} = \mathrm{FoM} - 20\log_{10}\left(\frac{\mathrm{FTR}}{10}\right), \tag{6}$$

is also reported, where FTR is the fractional tuning range in percent.

III. Simulation Results

A. Tuning Range and Frequency Coverage

The eight bank codes tile the band continuously from 5.226 GHz to 5.855 GHz, a span of 629 MHz corresponding to a fractional tuning range of 11.35 %. No dead zones occur at any code transition. The smallest band-to-band overlap is 34 MHz, which provides margin against process, voltage and temperature variation. The simulated band edges are listed in Table I. K_{VCO} falls monotonically from 102 MHz/V at the highest band to 81 MHz/V at the lowest, a total spread of 21 MHz/V, that is $\pm 11.5\,\%$ about the mean value, in agreement with the prediction of (3). The absolute value of K_{VCO} remains below 105 MHz/V across the entire tuning range.

B. Phase Noise, Output Amplitude and Figure of Merit

At the lowest band the oscillator achieves a phase noise of -116.4 dBc/Hz at a 1 MHz offset from a carrier of approximately 5.23 GHz, as shown in Fig. 3. The steady-state differential output is a clean sinusoid of amplitude ± 1.25 V, that is 2.5 V peak-to-peak differential, corresponding to a single-ended excursion of approximately ± 0.625 V about the common-mode level and therefore lying comfortably within the 1.8 V supply without clipping (Fig. 4). The oscillator core draws 3.57 mA, giving a DC power of 6.43 mW.

TABLE I
Simulated frequency coverage and tuning sensitivity of the eight switched-capacitor bank codes.

Bank code	Lower edge (GHz)	Upper edge (GHz)	Span (MHz)	K_{VCO} (MHz/V)
000	5.709	5.855	146	102
001	5.649	5.791	142	99
010	5.550	5.686	136	95
011	5.495	5.629	134	93
100	5.406	5.530	124	86
101	5.356	5.478	122	85
110	5.273	5.390	117	82
111	5.226	5.342	116	81

Substituting these values into (5) yields a figure of merit of -182.7 dB, and a tuning-range-normalized figure of merit $\mathrm{FoM_T}$ of -183.8 dB.

C. Effect of Tail Current on Band Overlap

An effect worth recording is that the tail current does not act only on amplitude and phase noise: it also couples into the tuning map. Reducing the core current lowers the tank swing, which changes the average effective capacitance presented by the accumulation-mode varactors over an oscillation cycle, and therefore shifts both the band edges and the varactor span. Lowering the core current from 3.57 mA to 2.1 mA was found to collapse the worst-case band overlap from 34 MHz to approximately 7 MHz, which is too small to survive process and temperature variation. The higher current was retained for robustness, accepting the corresponding penalty in DC power. This coupling means that bias current and bank sizing cannot be optimized independently.

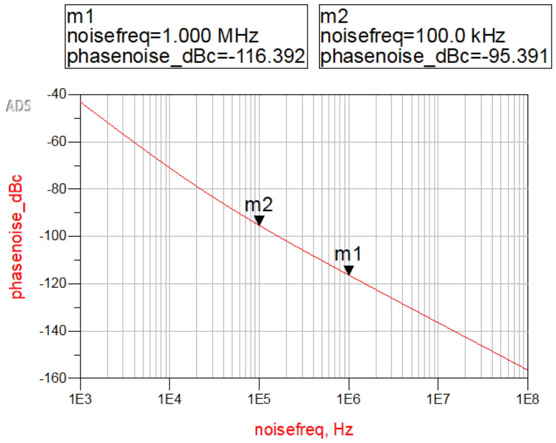

Fig. 3. Simulated phase noise at the lowest frequency band, showing −116.4 dBc/Hz at a 1 MHz offset.

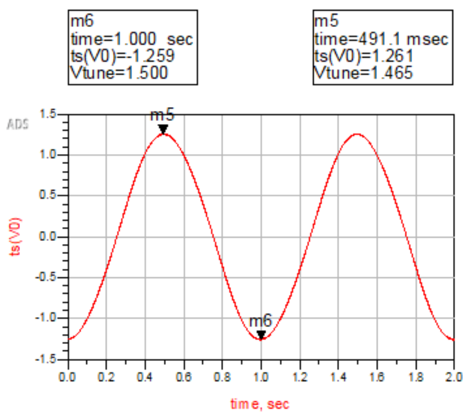

Fig. 4. Steady-state differential tank voltage, showing a clean sinusoidal waveform of amplitude ±1.25 V.

D. Performance Summary

Table II summarizes the simulated performance of the oscillator.

E. Preliminary Layout

The scope of this study is schematic-level design and characterization. A preliminary layout of the core was nevertheless drawn in Cadence Virtuoso with the TSMC 0.18 μm process design kit (Fig. 5). Due to the time constraint, *DRC and LVS are incomplete, no parasitic extraction has been performed, and no post-layout performance is claimed.*

IV. Comparison and Conclusion

A complementary cross-coupled LC voltage-controlled oscillator with a 3-bit switched-capacitor bank has been designed and characterized at the schematic level in a 0.18 μm RF CMOS technology. The oscillator covers 5.226–5.855 GHz, a fractional tuning range of 11.35 %, achieving a phase noise of −116.4 dBc/Hz at a 1 MHz offset and a figure of merit of −182.7 dB, while holding K_{VCO} within ±11.5 % of its mean across the whole tuning range.

Table III compares the oscillator with recently published LC-VCOs in comparable technologies. Direct comparison between oscillators is complicated by the fact that phase noise, power consumption, tuning range and carrier frequency can be traded against one another; the figure of merit normalizes

TABLE II
Summary of the Simulated Performance of the Oscillator.

Parameter	This work
Technology	TSMC 0.18 μm RF CMOS
Supply voltage	1.8 V
Topology	Complementary cross-coupled LC, 3-bit switched-capacitor bank
Tuning range	5.226–5.855 GHz (629 MHz)
Fractional tuning range	11.35 %
K_{VCO}	81–102 MHz/V (±11.5 %)
Phase noise @ 1 MHz offset	−116.4 dBc/Hz
Differential output swing	±1.25 V (2.5 V peak-to-peak)
Core current	3.57 mA
DC power	6.43 mW
Figure of merit (FoM)	−182.7 dB
$\mathrm{FoM_T}$	−183.8 dB

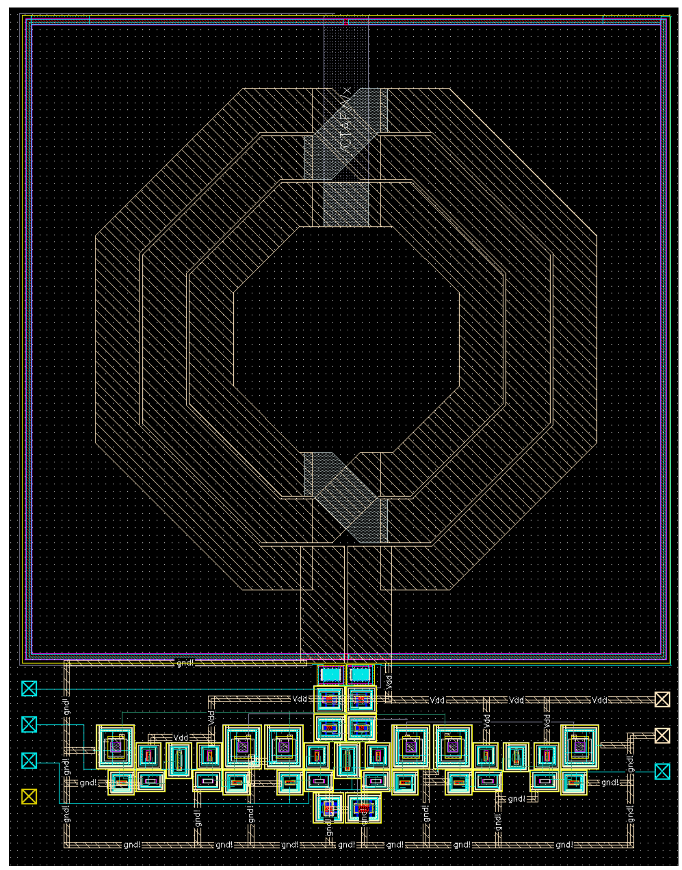

Fig. 5. Preliminary layout of the oscillator core, drawn in Cadence Virtuoso. The view is unverified: DRC and LVS are incomplete and no parasitic extraction has been performed.

TABLE III
COMPARISON WITH RECENTLY PUBLISHED LC VOLTAGE-CONTROLLED OSCILLATORS.

	Tech. (μm)	Supply (V)	Freq. (GHz)	FTR (%)	PN @ 1 MHz (dBc/Hz)	Power (mW)	FoM (dB)	ΔK_{VCO} (%)
This work	0.18	1.8	5.23–5.86	11.35	−116.4	6.43	−182.7	±11.5
[4]	0.18	1.8	4.10–5.70	32.65	−118.0	6.12	−182.4	—
[5]	0.18	1.8	4.24–4.99	16.25	−113.9	6.30	−179.2*	±10
[6]	0.18	1.8	4.39–5.26	18.00	−113.7	9.70	−180.0	±9.56

*Not reported by the authors; computed from (5) using the published phase noise, carrier frequency and DC power.

the first three but not the fourth. Within that limitation, the present design is competitive in figure of merit with published work in the same process and frequency band, and compares favourably in gain flatness.

Future work comprises experimenting with transformer-coupled topologies to reduce flicker-noise upconversion, lengthening the tail devices, which are presently at minimum channel length and constitute a known flicker-noise upconversion path, designing the buffer required to drive the subsequent mixer stage, and completing the layout of Section III-E through DRC, LVS and post-extraction verification ahead of integration into a phase-locked loop.

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REFERENCES

[1] K. H. Koo, S. W. Yoon, and C. W. Kim, "A 34% tuning range CMOS VCO with linear frequency tuning," in *Proc. IEEE Radio and Wireless Symp. (RWS)*, 2009.

[2] A. Hajimiri and T. H. Lee, "A general theory of phase noise in electrical oscillators," *IEEE J. Solid-State Circuits*, vol. 33, no. 2, pp. 179–194, Feb. 1998.

[3] B. Razavi, *Design of CMOS Phase-Locked Loops: From Circuit Level to Architecture Level*. Cambridge, U.K.: Cambridge Univ. Press, 2020.

[4] C. C. Wang and Y. F. Kuo, "5 GHz automatic amplitude calibration VCO using synchronous clock temperature compensation," in *Proc. 21st Int. SoC Design Conf. (ISOCC)*, 2024.

[5] Y. C. Chiang, T. H. Tseng, and C. T. Chen, "A 4.24–4.99 GHz complementary LC VCO with low K_{VCO} variation in 180-nm CMOS," in *Proc. IEEE 2nd Int. Conf. Consumer Technology (ICCT-Pacific)*, 2026.

[6] Y. J. Moon, Y. S. Roh, C. Y. Jeong, and C. Yoo, "A 4.39–5.26 GHz LC-tank CMOS voltage-controlled oscillator with small VCO-gain variation," *IEEE Microw. Wireless Compon. Lett.*, vol. 19, no. 8, pp. 524–526, Aug. 2009.